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3.3.1 Number of research papers published per teacher in the Journals as notified on UGC CARE list during the Academic Year 2021

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Detection of Online Counterfeit Reviews Based On Text Classification Using Machine Learning

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ABSTRACT Online product selling is growing very rapidly. After purchasing the online products, consumer gives their reviews about that products. Sometimes that review may be authentic or deceptive. The artificial intelligence-based technique is capable to find the various predictions in the different application. Many researcher works on this domain earlier but most of them took nominal number of reviews and that too was hotel reviews, trip advisor for their proposed work. This research has large number of reviews for experimental purpose. As we all know that model produce more accurate results if we train with huge amount of data. This paper presents the machine learning techniques for fake review detection of online products using text classification. The various machine learning algorithms were implemented and the obtained results compared using accuracy, f1-score, classification error etc.

Keywords: Machine Learning, Detection, Fake Reviews, Model Accuracy, Error rate.

1. INTRODUCTION

Detecting fake ranking persons in any online e-commerce application reviews play a very important role. Now a day's people are aware with the fraudulent happening through the internet that's why they see each and every information provided in the sites and gives important to that information available before purchasing. So many people intentionally take advantage of this to get gain over the sites and they try to mislead through reviews and ratings. Websites which provide genuine products and services are mostly reliable and the reviews and rating given by the customers are worthful. So, online purchasing and services provided by the organization through online mode to the customer plays very important role and their feedback also. Some people give true feedback some mislead it by false reviews and few writes neutral. This work is aimed to identify that a reviews are fake or genuine.

In the present scenario, customer depends fully on reviews given by others and trust them. Since, these reviews played important role as it can be fake, genuine or neutral. Reviews manipulated for positive or negative opinions are known as opinion spam. Now a days with deceptive opinion spam are big threat to both customer and organizations. Many organizations and customer intentionally plan and paid to write such untruthful

PROCESS OF MANUFACTURING & COMPOSITE MATERIAL APPLICATION

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ABSTRACT:- The composite material (composition material) is a materials make from two, three or more constituent materials with significant different physical, metallurgical or chemical properties that, when combined, produce a material with attribute different from the individual components. in this paper presents precise manufacturing process' techniques of composite materials and them applications or uses. in this paper presented some manufacturing processes of composites, as filament winding, laminating , resin transfer molding, pultrusion, and them large applications in automobile and aeronautics etc.

Keywords: composite material, automobile, natural fibers.

A. INTRODUCTION

Composite material have obtain a widely uses in all different ranges of modern materials as aeronautic, automotive, sports parts and therapeutic uses. the general definition, the composite material has more category, and ones of them can be as a material composed by the combination of two, three or more materials. a strengthen (reinforce) element and a compatible resin epoxy (matrix) to got specific feature and properties . The roles of matrix in composite materials are to give formation to the composite part, protect the strengthen to the environment, transfer loads to reinforcements and strength (soundness) of material, related with reinforcements. The focus of reinforcements in composites are to become better, stiffness , mechanical properties and metallurgical property , dominate other properties as volume of thermal extension, conductivity and thermal transports. [1]

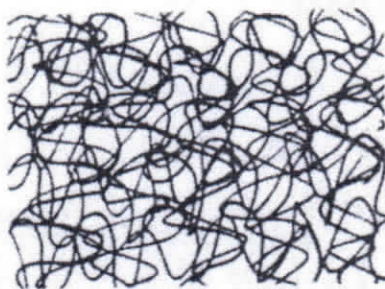
The comparison between composites and metals, composites materials advantages are followings:

- (i) Internal energy , Lightweight ,storage and release.
- (ii) Good specific stiffness and strength,
- (iii) Good dumping,
- (iv) Easy bondable,
- (v) Easy moldable to complex forms
- (vi) Good fatigue resistance,
- (vii) Low electrical conductivity and thermal expansion,
- (viii) Low radar visibility,
- (ix) Lightweight

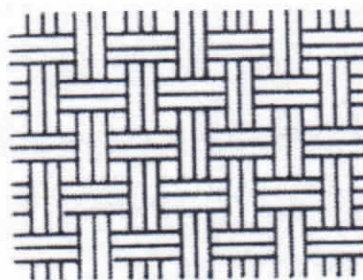
Some disadvantages of composites are:

- (i) materials Cost comparatively high
- (i) Long development time,
- (i) Temperature limits,

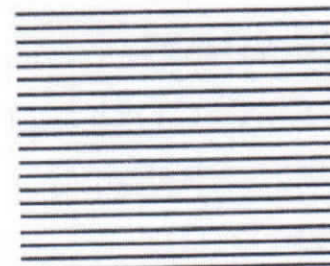
All these things made those composites to change more and more the metals, specials in marines, automotive, aircrafts, etc. common type of composite material is glass fiber, which is very good strength, fracture readily if notched, and at compression, it buckles easily. It can be protected from destruction by encapsulating the glass fiber in a resin matrix, which transfers apply loads in the unified fibers, due to used it both for compression and tension [2][3]



Continuous Strand Mat (CSM)



Woven



Unidirectional Mat (UDM)

Fig 1 form of glass fibre [1]



Analytical and Experimental Investigation of a Plate Fin Heat Exchanger at Cryogenics Temperature

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Keywords:

plate fin heat exchanger, effectiveness, friction factor, cryogenics

ABSTRACT

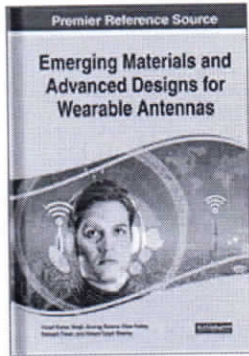
Plate-fin heat exchangers provide a broad range of applications in many cryogenic industries for liquefaction and separation of gases because of their excellent technical advantages such as high effectiveness, compact size, etc. Correlations are available for the design of a plate-fin heat exchanger, but experimental investigations are few at cryogenic temperature. In the present study, a cryogenic heat exchanger test setup has been designed and fabricated to investigate the performance of plate-fin heat exchanger at cryogenic temperature. Major parameters (Colburn factor, Friction factor, etc.) that affect the performance of plate-fin heat exchangers are provided concisely. The effect of mass flow rate and inlet temperature on the effectiveness and pressure drop of the heat exchanger are investigated. It is observed that with an increase in mass flow rate effectiveness and pressure drop increases. The present setup emphasizes the systematic procedure to perform the experiment based on cryogenic operating conditions and represent its uncertainties level.

1. INTRODUCTION

Compacts heat exchangers provide a broad range of applications such as aerospace, automobiles, radiators, condensers, evaporators, etc. To enhance the rate of heat transfer, different types of fins (plain, offset, perforated, louver, wavy, etc.) were used in heat exchangers. Starting earlier from 1942, Norris and Spofford [1] had reported the first experimental results on offset strip fin. Three types of offset fins were tested and draw out the effect of heat transfer coefficient, friction factor, and Colburn factor by varying its length, thickness and pitch of fins. Joyner [2] conducted an experimental analysis to measure total heat transfer and pressure drop of five serrated fins of different lengths. The results illustrated the local heat-transfer coefficient, Nusselt number, Reynolds number, and friction factor. Manson [3] predicted the first empirical correlation on which one equation was for heat transfer over the entire range of conditions was proposed. He concluded that the friction factor data was required for a separate correlation below and above the transition Reynolds number (3500). However, the database comprised of different geometries such as louvers fins, finned flat tubes, and offset strip fins. Kays and London [4, 5] carried out the experimental investigations for offset fin geometry in two parts. In the first part, a detailed descriptions of the experiment apparatus and method for test data analysis were provided whereas in the second part the correlation for Colburn factor (j) and friction factor (f) for two different offset strip fins cores were proposed. Later on, the j and f design data of three offset plate-fin was summarized [6].

Further, Kays [7] explored the research on analogous geometry along with six more heat transfer surface that was the first effort for calculating the overall efficiency of the finned passage. Briggs and London presented design data of j and f for eleven compact plate-fin surfaces (five surfaces for offset rectangular-fin type and six for plain triangular-fin type) [8, 9]. The cores were made of aluminum and the rest part were of an alloy of stainless steel and nickel that was suitable for high-temperature applications. London and Shah [10] extended the work on heat transfer and flowed friction design data of nine offset rectangular plate-fin surfaces two of them was made up of stainless steel. Voronin and Dubrovsky [11] developed a correlation for friction factor and Nusselt number. They found that the flow separation due to interruption played a significant role. Wieting [12] set up the statistical relationship between the variable from earlier experimental heat transfer and flow friction data of an offset fin for the plate-fin heat exchanger. By using these statistical relationship, untested offset fin geometries can be predicted realistically and accurately within the parametric range of previously tested data. The connection between the various data based on information collected from 22 balance-offset fins and provided the correlation for both the range laminar and turbulent regions furthermore created connections for discriminating Reynolds number. Mochizuki and Yagi [13] experimentally analyzed the optimized value of fin length. The study demonstrated the optimized value of the fin length for better performance of heat exchanger (seven aluminum fin cores).

Sparrow and Hajiloo [14] demonstrated the uses of the naphthalene sublimation technique to achieve the heat transfer



Analysis and Design of Solar Dryer for Forest Produce of Bastar Region in State of Chhattisgarh, India

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Abstract

In the Chhattisgarh state of India, the Bastar region is famous for its various forest produce collected by local inhabitants. These collected forest produce are dried in open sun drying. This chapter presents the design and analysis of solar dryer which is simple in construction and a low-cost dryer. The construction of the solar dryer will be made of galvanized mild steel sheet absorbing the sun's radiation. The hot air will be taken out through a wind turbine ventilator at the top to maintain the required humidity level in the solar dryer. The analysis shows that the temperature inside the solar dryer is evenly distributed. The suggested solar dryer will reduce the drying time of the forest produce.

Chapter Preview

Top

Criteria Of Solar Dryer For Bastar Region

For the selection of solar dryer the main selection criteria are (Visvale, 2012):-

- Ease of design
- Can handle small and medium quantity of forest produce
- Assemble with easily available materials
- No use of electricity
- Less Cost

The selected design is further modified on the Ooi's suggested model (Ooi, 2006). It is basically a box dryer. To control the hot air flow inside the dryer a roof top turbine ventilator is used. The dryer is further analyzed and found the even temperature and velocity distribution in the dryer to maintain the desired humidity level is achieved through this. This will further improve the texture and quality of the forest produce. The Ooi (2006) incorporated solar drying into a box dryer with area 2.4 m X 2.4 m for bed depth of 0.3 m. This box dryer outer surface is covered by corrugated sheets. The selected dimension for analysis on Ooi's box dryer are kept 6 feet (1828.80 mm) in length and width is 3 feet (914.40 mm) and the upper height is kept as 2.62 feet (800mm) and lower height is 1.53 feet (467.19 mm) at maximum [fig. 1]. This will reduce the dimensions of Ooi's suggested model. The slant of the solar dryer will be at an angle of 170°. Figure shows the wire frame diagram of the suggested Solar Dryer. For the study the longitude is taken as 81.39909° and latitude is taken as 18.9739° [Fig. 3]. These coordinates are of Dantewada city.

Fabrication, Prototype, and Testing of Acetylene Generator for Low-pressure Oxyacetylene Welding.

- **Source:** Turkish Online Journal of Qualitative Inquiry . 2021, Vol. 12 Issue 3, p5268-5271. 4p.
- **Author(s):** Tapas, Vikrant; Sahu, Prem Shankar; Chandrakar, Haldhar Ram
- **Abstract:** The Calcium Carbide based Low-Pressure Acetylene Generators are used extensively in India. The low-pressure Acetylene generator is used with an Industrial oxygen cylinder for oxy acetylene welding. These Calcium Carbide-based Low-Pressure Acetylene generators are manufactured by the welders or purchased from the market. In these used acetylene generators fewer safety measures are followed. They are unsafe due to their capacity, water level indication in the tank, and the bottom tank inclination. To overcome this gap in these drawbacks a prototype is developed and named as SHASHI (Stimulated Heated Acetylene Safe High Injection) is based to overcome these drawbacks. The dimension, temperature variations details are discussed in detail in this research paper. The prototype is tested for maximum 1.07 kg/cm² and at maximum temperature 72°C. The shared information will certainly help for the researchers, welders and other associated members to further improvement in standardization in this field.
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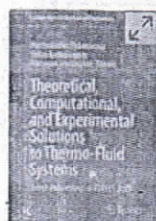


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Theoretical, Computational, and Experimental Solutions to Thermo-Fluid Systems pp 239–252

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Effect of Porosity on the Performance of an GM-Type DIPTR Using CFD

[Pankaj Kumar](#) , [Ajay Kumar Gupta](#), [Sanjay Kumar Gupta](#) & [R. K. Sahoo](#)

Conference paper | [First Online: 10 March 2021](#)

449 Accesses

Part of the [Lecture Notes in Mechanical Engineering](#) book series (LNME)

Abstract

The present research goals to establish a better reliable as well as stable simulation of pulse tube refrigerator (PTR) by commercially available computational fluid dynamic (CFD) software. The operation of double inlet PTR (DIPTR) is mostly depending upon a regenerator. Regenerator plays a

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Influence of dispersion technique/time on dispersion stability, aspect ratio and morphology of multi-walled carbon nanotubes

Sourabh Kumar Soni, Vedansh Tody and Benedict Thomas

Published Online: 21 Jun 2021



Abstract

Current interest in nanoscience and nanotechnology has motivated a remarkable research effort in carbon nanotubes (CNTs), which has addressed a variety of issues and produced several new findings. The uniform and stable dispersion of CNTs in solvents/nanocomposites is often a necessity for their effective use in advanced materials. In this study, the dispersion states of multiwalled carbon nanotubes (MWCNTs) in numerous solvents were analysed by UV-vis spectroscopy and visual observations (photographic images). Furthermore, the influence of sonication and ball milling time on the MWCNT morphology and dispersion was analysed through scanning electron microscopy (SEM) and transmission electron microscopy (TEM). It was found that the MWCNT was highly dispersible in DMF, isopropanol, chloroform and acetone, and the dispersion state of MWCNT in these solvents can be classified as dispersed. The obtained results indicate that the oxidation incorporates oxygen-containing groups on the surface of MWCNTs, which establish the electrostatic stability needed for long-period stable dispersion in solvents. Morphological investigation of MWCNTs shows less than 20 min of dispersion is insufficient to achieve untangled CNTs with good dispersion, while 20-60 min of dispersion demonstrates CNTs bonding, intersections, junctions and number of defects on MWCNTs due to high energy involved. Apart from this, significant MWCNT shortening has been observed for the dispersion period of 90 min-6 h. The investigation of the dispersibility of MWCNTs in different solvents and the effect of sonication and ball milling time on the MWCNT

Inventory Control Approach through Application of ERP Package in Steel Industry

Atish Goswami¹, Pradeep Agrawal²

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Abstract - Enterprise Resource Planning, or ERP, is about sharing information within the organization which generates, accesses, and relies on information to get the job done, whether that information is customer data, inventory details, job information, or financial statistics. Among departments such as finance, marketing, sales, customer service, and more, it can be a challenge to keep information accurate and up-to-date when each party is working its bulk with its data. The latest Enterprise Resource Planning modules are naturally interconnected and quickly spread across the world of data transfer, their use is better not only for active organizations and leads to similar growth. Since the latest ERP packages contain different modules, they serve as an advantage to the growth of any major.

The purpose of the study is to use the leading ERP Practices in the form of the latest SAP package namely SAP S4 HANA which will address the needs and requirements of the organization. The best part of this ERP solution is that it can find solutions for all companies with dedicated modules such as Production Planning Module, Material Management Module, Financial Planning Module, Human Resources Module, Account Module, Sales and Marketing, Procurement Management, Customer Module, Consumer Module Production etc. Communication created as a result of communication between all steel industry departments thereby reduces asset seizures, excessive financial burden, duplicate backups ordered, Use of immature goods with the same product value, Efforts required to complete the process according to their performance, efforts to simplify order from adoption its to the end.

My research of the SAP S4 HANA application in the steel industry in the production planning module emphasizes mainly on various aspects such as asset management, an inexpensive system by entering the right information at the right time in the most advanced ERP Package. SAP S4 HANA with new ideas discussed with professionals and the team I work with to support the organization meet the needs of an effective way to work with competent staff.

Key Words: Enterprise resource planning, Material requirement planning, manufacturing resource planning, Systems application and products (SAP), Master data management etc.

1. INTRODUCTION:

Enterprise Resource Planning (ERP) systems are at the forefront of these company-wide IT solutions these days.

Despite the difficulties in implementation, in large multinational corporations ERP systems have actually become the leading standard replacement for legacy solutions installed in each business unit separately (Shanks, 2000, quoted in Schlichter & Kraemmergaard, 2010). ERP systems have been in use for at least 20 years, however the implementation of the system is so complex that it comes with high costs and high failure rates. Despite the technical difficulties of ERP systems implementation is often accompanied by changes in business processes and organizational changes. The results of the project influence the daily work of the whole project. An example of the expected changes is the introduction of new operating procedures and the transformation of the daily responsibilities of employees. Issues such as resistance to change arise and need to be addressed. All of these factors summarize the complexity and are important for the future of a company project that represents many challenges for its project managers.

Business service planning uses multi-module software to improve the business performance of the internal process.

ERP packages were mainly targeted at the manufacturing industry and mainly included the planning and management activities of a core business.

Since its inception by the Gartner group they are pioneers of the ERP concept and have officially started practicing it in the early 90s.

1.1.1 Inventory Control Packages:-

ERP modules were previously known as Asset Management Packages in the Early 60s Containing only the performance of list control packages and are restricted to storage and local assets.

1960s Asset Management Packages

1970s Material Requirements Planning (MRP)

1980s Manufacturing Resources Planning (MRP II)

1990s Enterprise Resource Planning

Extended ERP of the 2000s

Table 1:- Overview of ERP emergence

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Published: 05 November 2021

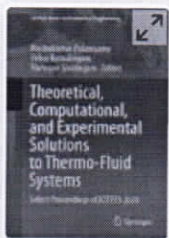
Microstructure and Mechanical Characterization of Al₂O₃/Graphene Reinforced Al6061 Based Hybrid Nanocomposites

[Sourabh Kumar Soni](#), [Daksh Ganatra](#), [Parth Mendiratta](#), [CH. S. K. Akhilesh Reddy](#) & [Benedict Thomas](#) 

Metals and Materials International **28**, 545–555 (2022)**441** Accesses | **15** Citations | [Metrics](#)

Abstract

The present work deals with the microstructure and mechanical characterization of the Al6061 based nanocomposites (2 wt% Graphene and 2 wt% Al₂O₃) and hybrid nanocomposites (2 wt% Al₂O₃ with 0.5, 1, 1.5, 2 and 2.5 wt% of Graphene) processed through ultrasonic-assisted melt-stirring technique. The combined effort of reinforcement processing (ball-milling) and ultrasonic-assisted melt stirring approach has been adopted for fabricating hybrid nanocomposites with uniformly dispersed ceramic nanoparticles and carbon nanomaterial. The grain



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Numerical Analysis of Pulse Tube Cryocooler and Optimization Using Taguchi Method

[Ajay Kumar Gupta](#) , [Pankaj Kumar](#)  & [R. K. Sahoo](#)

Conference paper | [First Online: 10 March 2021](#)

476 Accesses

Part of the [Lecture Notes in Mechanical Engineering](#) book series (LNME)

Abstract

The present work emphasizes the contribution in the development of software of pulse tube refrigeration. The process included the numerical modelling and optimization of dimensional parameter of pulse tube refrigerators on considering its performance parameters. In the development of these systems, as

Cascaded H Bridge 17 Level Inverter with PWM Technique

ROHIT KUMAR OBEROY¹, DR. PADMINI SHARMA²

^{1,2} Chhatrapati Shivaji Institute of Technology, Durg

Abstract- A Multilevel inverter is a very important tool for PV Array and Electric Vehicle etc. in Modern ERA Because the non-renewable energy is important for our future. The non-renewable energy is reducing the pollution and carbon content in the environment and also reducing the fossil fuel requirement for transportation and producing electricity. The multilevel inverter can provide higher level of sinusoidal output with less total harmonic distortion with the help of pulse width modulation. Different type of Pulse width modulation like POD APOD etc. can be implemented. 17 level inverters with the use of Phase Opposition Disposition pulse width modulation (POD-PWM) technique with 16 carrier signals with cascaded inverter with 16 IGBT switches is proposed. All the carrier signals are above X-axis are out of phase with the below signal by 180°. Total harmonic distortion of load voltage is 7.26% and load current is 3.09% obtained. This inverter can be use with solar photo voltaic cell. By the use of this inverter the higher-level output and less harmonic distortion output obtained. By the use of this output the electrical equipment is in very safe condition.

Indexed Terms- Multilevel inverter, PWM, POD, Total Harmonic distortion,

I. INTRODUCTION

The technique for the half-bridge inverter is optimized subjected to the constraint of switching frequency of the SCR's, used the concepts of modern control theory. Variable-frequency variable voltage sinusoidal output in three phase inverters was possible by employing the techniques developed in [1] new neutral-point-clamped pulse width modulation (PWM) inverter composed of main switching device which operated as switches for PWM and auxiliary switching devices to clamped the output terminal potential to neutral point potential had been developed in [2] A new commutation circuit, especially suitable for pulse

width-modulated (PWM) inverters, was described. The circuit did not require recharging of a commutation capacitor and was therefore ready for commutation at any time and this improved the Circuit efficiency in [3] a generalization of the PWM "subharmonic" method to control single phase or three phase multilevel voltage source inverters (VSI) is reveal in the [4-8] Multilevel Inverter was a key technology and plays critical role in AC motor drives, uninterruptible power supplies, high-voltage DC power transmission, flexible AC transmission systems, static var compensators, active filters, electric and hybrid electric vehicles and integration and utilisation of renewable energy sources [9,10]. In the field of high-power medium voltage DC/AC conversion, MLI is getting tremendous popularity together in terms of topology & control scheme because of its decent power quality, less total harmonic distortion (THD), reduced voltage tension across the switches, good electromagnetic compatibility, fewer switching losses and dv/dt stress. However, MLI possess some disadvantages, that was, to rise output levels, the number of semiconductor switch necessity along with peripherals devices like gate driver circuit, protection circuit & heat sink rises. Increased device count makes overall system complex, bulky & costly and reduced the efficiency and reliability of the converter.

Traditionally, MLIs are classified as cascaded H bridge (CHB), flying capacitor (FC) & neutral point clamped (NPC). In past few decades, most of the literatures printed showed the study on CHB, FC and NPC topologies with respect to their corresponding advantages and disadvantages [10] A CHB MLI was composed of several H bridge cell & isolated DC source. On the basis of voltage magnitude DC source, CHB was classified as asymmetric and symmetric configuration. In symmetric configuration, magnitudes of DC sources were equal ($V_1 = V_2 = V_3 \dots$), whereas in asymmetric configuration

H Bridge 3 Level Inverter with Pulse Width Modulation Technique

ROHIT KUMAR OBEROY¹, DR. PADMINI SHARMA²

^{1,2} Chhatrapati Shivaji Institute of Technology Durg

Abstract- A Multilevel inverter is a very important tool for PV Array and Electric Vehicle etc. in Modern ERA Because the non-renewable energy is important for our future. The non-renewable energy is reducing the pollution and carbon content in the environment and also reducing the fossil fuel requirement for transportation and producing electricity. The multilevel inverter can provide higher level of sinusoidal output with less total harmonic distortion with the help of pulse width modulation. Different type of Pulse width modulation like POD APOD etc. can be implemented. 3 level inverters with the use of Phase Opposition Disposition pulse width modulation (POD-PWM) technique with 2 carrier signal with inverter with 4 IGBT switches is proposed. All the carrier signals are above X-axis are out of phase with the below signal by 180°. Total harmonic distortion of load voltage is 35.45 % and load current is 35.51 % obtained. This inverter can be use with solar photo voltaic cell. By the use of this inverter the higher-level output and less harmonic distortion output obtained. By the use of this output the electrical equipment is in very safe condition.

Indexed Terms- Multilevel inverter, Inverier, POD, PWM, Total Harmonic Distortion

I. INTRODUCTION

The technique for the half bridge inverter is heightened subjected to the limitation of switching frequency of SCR's, used the perceptions of modern control theory. Variable-frequency variable voltage sinusoidal output in three phase inverters was probable by using the methods developed in [1] new neutral-point-clamped pulse width modulation (PWM) inverter collected of core switching devices which operated as switches for PWM & auxiliary switching devices to clamped the harvest terminal potential to neutral point potential had been developed in [2] A novel commutation circuit, specially appropriate for

pulse width-modulated (PWM) inverters, was described. The circuit did not require recharging of a commutation capacitor and was therefore ready for commutation at any time and this improved the Circuit efficiency in [3] a generalization of the PWM "subharmonic" method to control single-phase or three-phase multilevel voltage source inverters (VSI) is revel in the [4-8] Multilevel Inverter is a key technology & acting crucial role in AC motor drives, uninterruptible power supplies, high-voltage DC power transmission, flexible AC transmission systems, static var compensators, active filters, electric and hybrid electric vehicles and integration and utilisation of renewable energy sources [9,10]. In the field of high-power medium voltage DC/AC translation, MLI is reception marvelous popularity together in terms of topology & control scheme because of its upright power quality, fewer total harmonic distortion (THD), condensed voltage pressure across switches, good electromagnetic compatibility, less switching losses & dv/dt stress. However, MLI possess almost drawbacks, that is, to growth output levels, the number of semiconductor switch requirement sideways peripherals devices like gate driver circuit, protection circuit & heat sink increases. Improved device count makes total system composite, bulky & costly & reduces efficiency and the reliability of the converter.

Traditionally, MLIs are classified as cascaded H bridge (CHB), flying capacitor (FC) and neutral point clamped (NPC). In past limited decades, utmost of the literatures published displays study on CHB, FC and NPC topologies with reverence to their corresponding benefit and drawbacks (8) A CHB MLI is collected of several H bridge cell & isolated DC source. On the basis of voltage magnitude DC source, CHB is classified as symmetric and asymmetric configuration. In symmetric configuration, extents of DC sources are like ($V_1 = V_2 = V_3 \dots$), whereas in asymmetric configuration magnitudes of DC sources are not alike

Stock Market Prediction Using LSTM Techniques in Machine Learning

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Abstract

One of the most multifaceted Artificial Intelligence Machine Learning issues is the offer worth forecast. It relies upon an assortment of elements that influence gracefully and request. This paper dissects various techniques for determining the future stock cost and gives a model utilizing a pre-constructed model that is adjusted to the Indian financial exchange. Costs of stocks are portrayed by time-arrangement information, and neural organizations are prepared to take in the examples from patterns in the current information. In this exploration, we additionally look at between aftereffects of a straightforward single LSTM model, a stacked LSTM model adjusted to the Indian market by utilizing the Indian BANK stock value informational index, and a mixture model utilizing both stacked auto-encoders and estimation investigation.

Keywords:- Artificial Intelligence, Machine Learning, LSTM

I. Introduction

The stock market is basically an aggregation of various buyers and sellers of stock. A stock (also known as shares more commonly) in general represents ownership claims on business by a particular individual or a group of people. The attempt [3] to determine the future value of the stock market is known as a stock market prediction. The prediction is expected to be robust, accurate and efficient. The system must work according to the real-life scenarios and should be well suited to real-world settings. The system is also expected to take into account all the variables that might affect the stock's value and performance. There are various methods and ways of implementing the prediction system

like Fundamental Analysis, Technical Analysis, Machine Learning, Market Mimicry, and Time series aspect structuring. With the advancement of the digital era, the prediction has moved up into the technological realm. The most prominent and [3] promising technique involves the use of Artificial Neural Networks, Recurrent Neural Networks that is basically the implementation of machine learning. Machine learning involves artificial intelligence which empowers the system to learn and improve from past experiences without being programmed time and again. Traditional methods of prediction in machine learning use algorithms like Backward Propagation, also known as Back propagation errors. Lately, many researchers

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28. Exposure of PM_{2.5} and Carbonaceous Matter Amongst Rural Inhabitants in and Around Durg District of Chhattisgarh, India

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Abstract

This study was carried out in Durg District of Chhattisgarh, comprising of 83 villages. It was the need for monitoring and analysis of PM_{2.5} to perceive the scenario and exposure of particulates over region located in rural areas of agricultural state of Chhattisgarh, dominated with anthropogenic activities related to heavy biomass fuel burning. Three rural areas, Achhoti (AC), Anjora (AN), Chandkhuri (CH) were selected and monitoring of PM_{2.5} and carbonaceous matter were carried out during winter and summer seasons in 2014–2015. The average seasonal variations of PM_{2.5} concentrations for winter and summer seasons for all sampling locations were recorded in the following order; Chandkhuri (CH) $611.37 \pm 112.57 \mu\text{g m}^{-3}$ and $428.74 \pm 80.64 \mu\text{g m}^{-3}$ > Achhoti (AC); $533.13 \pm 98.5 \mu\text{g m}^{-3}$ and $380 \pm 74.19 \mu\text{g m}^{-3}$ > Anjora (AN); $524.36 \pm 99.42 \mu\text{g m}^{-3}$ and $317 \pm 59.77 \mu\text{g m}^{-3}$, respectively. As per the annual standards given by CPCB and USEPA, the concentration of

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